

Project 1 - Data Architecture & Modeling: European Football Results DW

LTAT.02.007 Data Engineering, University of Tartu, autumn 2026 · Group 14 · Author: Ron-Aran Paju (individual project) · Repository: <https://github.com/Aran211/football-dw>

1. Business brief

Objective. Build an analytics model that shows how squad market value and bookmaker expectations translate into actual results in Europe's top-5 football leagues (2023/24 – 2026/27).

Stakeholders. Club performance & recruitment analysts (value for money of a squad), sports-betting / trading analysts (how well the market prices matches), sports journalists and fans' media (data-driven stories on home advantage, referees, upsets).

KPIs. (1) *Points per €100 m of squad value* – efficiency of spending; (2) *Favourite hit rate* – % of matches won by the pre-match bookmaker favourite (and its complement, upset rate); (3) *Home advantage* – home win % and average home-minus-away goal difference.

Business questions (each has a demo SQL query, section 7):

- Q1 Which teams over- or under-perform relative to their squad market value in each season?
- Q2 How often does the bookmaker favourite win, and which league is the least predictable?
- Q3 How large is home advantage per league, and is it changing across seasons?
- Q4 Which teams convert chances best (goals per shot on target), and does it go with more points?
- Q5 Do referees differ in cards and fouls per match? (referees are published for England only)
- Q6 Did a team's results change after a coach change? (uses SCD2 history of `dim_team`)
- Q7 Which were the biggest upsets (winners with the longest pre-match odds)?

2. Datasets

Dataset	Content (grain)	Size	Update / access	Why
A. football-data.co.uk mmz4281/<season>/<div>.csv	One row per league match: date, kick-off, teams, full/half-time score, shots, shots on target, corners, fouls, cards, referee (England), xG (from 2026/27, most leagues), 1X2 odds of many bookmakers + market average, opening and closing	~1,750 matches per season across the 5 leagues (3×380 + 2×306) → ~5,500 rows for 3 full seasons + current; 100+ columns	CSV over HTTP, free, no key; updated during the season about twice a week	Results, match events and the market's expectation per match
B. Transfermarkt dataset (dcaribou/transfermarkt-datasets, Kaggle & GitHub)	12 joinable tables; used: <i>clubs</i> (stadium, league), <i>players</i> (current club, position, value), <i>player_valuations</i> (one row per valuation event), <i>games</i> (manager of each club per match → coach history)	players 50,000+ rows × 20+ cols; valuations 650,000+; games 88,000+ × 20+ cols; clubs 790+	CSV files, weekly pipeline – paused since July 2026 (valuations end 2026-06-12)	Squad market value and team attributes (coach, stadium) the match files lack

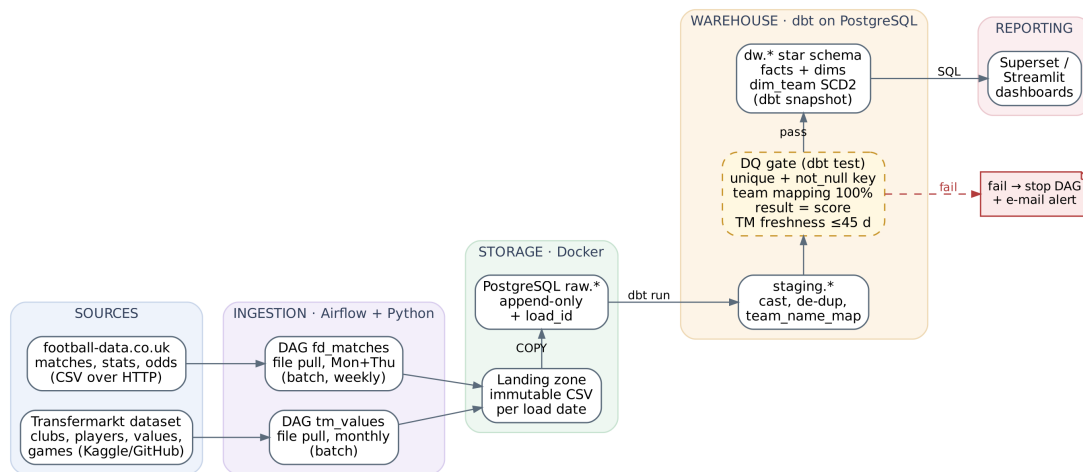
Sources. [A] football-data.co.uk, <https://www.football-data.co.uk/data.php> (key: /notes.txt); [B] D. Cereijo, transfermarkt-datasets, <https://github.com/dcaribou/transfermarkt-datasets>, Kaggle: <https://www.kaggle.com/datasets/davidcariboo/player-scores> (data © Transfermarkt.com).

Combining the datasets. Sources share no key: team names differ ("Man United" vs. "Manchester United", "Ath Madrid" vs. "Atlético de Madrid"). A curated bridge *staging.team_name_map* (exact → fuzzy → manual match, reviewed flag) maps every football-data name to a Transfermarkt *club_id*; a DQ test blocks the load if any name is unmapped. Player valuations are aggregated per club and month-end into a squad-value snapshot, which is attached to each match as of the match date. **Risk:** because Transfermarkt updates are paused, 2026/27 matches use the June 2026 snapshot; this is exposed through *value_snapshot_date_key* and a freshness check rather than hidden; coach changes after July 2026 are captured once upstream resumes.

3. Tooling by lifecycle stage (tools covered in the course)

Stage	Tool	Use in this project
Ingestion	Python + Apache Airflow	DAGs download CSVs (requests / Kaggle API), write to landing zone, COPY into raw schema; retries, scheduling, alerting
Storage	Docker (volume) + PostgreSQL	Landing zone of immutable files per load date; PostgreSQL holds raw, staging and dw schemas
Transformation	dbt	Staging models (cast, de-dup, mapping), dbt snapshots for SCD2 <code>dim_team</code> , mart models for the star schema, dbt tests = DQ gate
Serving / BI	Superset, Streamlit	Dashboards for KPIs; Streamlit page for 'team vs. its value' exploration
Packaging	Docker Compose	One stack: postgres, airflow, dbt, superset – reproducible for Project 2
Not used	MongoDB, Neo4j	Both sources are already tabular; no document or graph workload needed

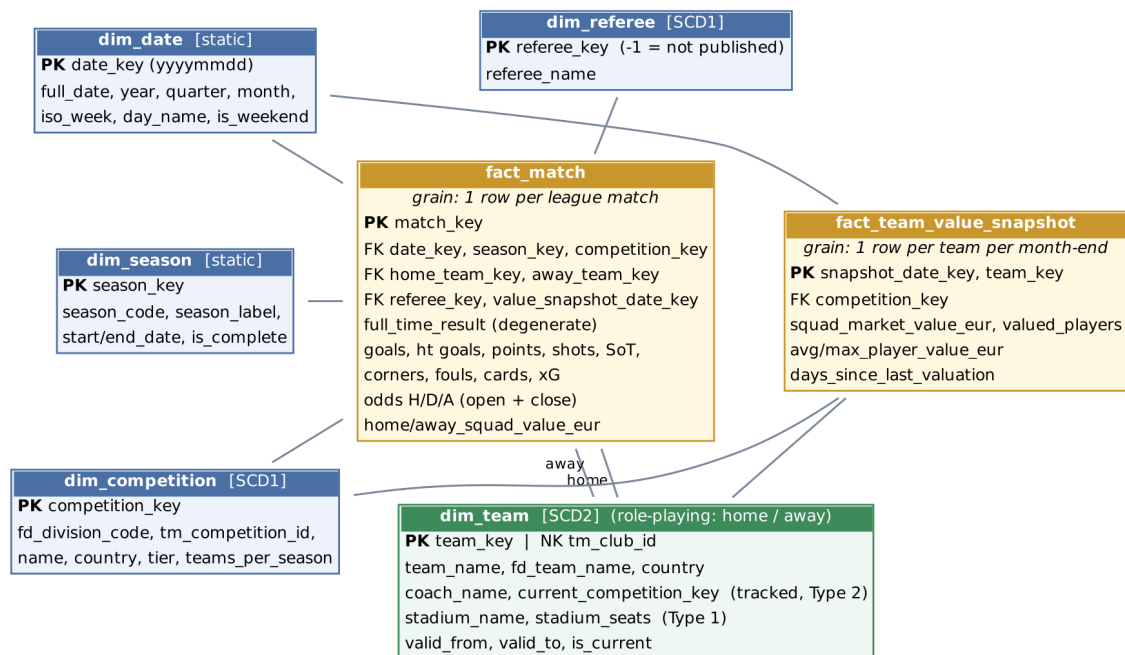
4. Data architecture



Flow step	Method	Frequency
Source → ingestion (matches)	Batch file pull over HTTP (current-season CSV per league); full history back-filled once	Mon + Thu 06:00 during season (matches are played weekends + midweek)
Source → ingestion (values)	Batch file pull via Kaggle API (clubs, players, player_valuations, games)	Monthly (1st day); source itself updates at most weekly
Landing → raw	COPY into append-only raw tables with load_id; files kept for replay	After each pull
raw → staging → dw	dbt run: incremental fact_match upsert on natural key, dbt snapshot for dim_team, full rebuild of small dims	Triggered by Airflow after each load
dw → reporting	Superset / Streamlit read dw.* directly	On demand

Data quality checks (run as dbt tests; *error* stops the DAG, *warn* alerts): uniqueness of (division, date, home, away) and not-null on mandatory fields [error]; every team name mapped to a Transfermarkt club [error]; result letter consistent with the score [error]; odds > 1.01 and shots on target ≤ shots [warn]; completeness – a finished season has $n \cdot (n-1)$ matches (380 / 306) [warn]; freshness – newest valuation ≤ 45 days old [warn, currently failing by design]. SQL in `sql/04_data_quality_checks.sql`.

5. Data model (star schema)



Grain. *fact_match*: one row per league match (one fixture between a home and an away team in one competition and season), as published by football-data.co.uk. It is a transaction fact; dim_team is used twice (role-playing home/away), result is a degenerate dimension. Goals, shots, cards and points are additive; odds are non-additive (averaged or converted to implied probabilities); squad value is semi-additive. *fact_team_value_snapshot*: one row per team per month-end – a periodic snapshot, because valuations arrive irregularly per player and analysts need a consistent squad value at a point in time.

Dimension	SCD type	Justification
dim_team	Type 2 (coach, league); Type 1 (stadium seats)	Coaches change and teams are promoted/relegated; Q6 and season comparisons need the version valid on match day, so changes add a new row (valid_from/valid_to/is_current). History is back-filled from the manager name per match (TM games) and the league per season (match files); new changes via dbt snapshot. Stadium capacity corrections are not analytically meaningful → overwrite.
dim_competition	Type 1	Reference data (codes, country, tier). A sponsor/name change should relabel all history, not split it.
dim_referee	Type 1	Only the name is known; corrections of spelling overwrite. Unknown member –1 for leagues without referee data.
dim_date, dim_season	Static	Generated calendars; values never change after creation.

6. Data dictionary (all columns and types; per-column descriptions also in docs/data_dictionary.md)

Table	Columns and data types	Description / notes
dw.fact_match	match_key BIGINT, date_key INTEGER, season_key SMALLINT, competition_key SMALLINT, home_team_key INTEGER, away_team_key INTEGER, referee_key INTEGER, value_snapshot_date_key INTEGER, kickoff_time TIME, full_time_result CHAR(1), home_goals SMALLINT, away_goals SMALLINT, home_ht_goals SMALLINT, away_ht_goals SMALLINT, home_points SMALLINT, away_points SMALLINT, home_shots SMALLINT, away_shots SMALLINT, home_shots_on_target SMALLINT, away_shots_on_target SMALLINT, home_corners SMALLINT, away_corners SMALLINT, home_fouls SMALLINT, away_fouls SMALLINT, home_yellow_cards SMALLINT, away_yellow_cards SMALLINT, home_red_cards SMALLINT, away_red_cards SMALLINT, home_xg NUMERIC(4,2), away_xg NUMERIC(4,2), odds_home NUMERIC(6,2), odds_draw NUMERIC(6,2), odds_away NUMERIC(6,2), odds_home_close NUMERIC(6,2), odds_draw_close NUMERIC(6,2), odds_away_close NUMERIC(6,2), home_squad_value_eur BIGINT, away_squad_value_eur BIGINT, source_file VARCHAR(20), load_id BIGINT	Transaction fact, 1 row per league match. Keys * key are FKs (home/away → dim_team version valid on match day; referee –1 = not published; value_snapshot_date_key = month-end of squad values used). full_time_result H/D/A = degenerate dim. *points 3/1/0 derived. xG NULL before 2026/27. odds_* = market-average decimal odds, pre-match and *close. *squad_value_eur = latest snapshot ≤ match date. source_file, load_id = lineage.
dw.fact_team_value_snapshot	snapshot_date_key INTEGER, team_key INTEGER, competition_key SMALLINT, squad_market_value_eur BIGINT, valued_players SMALLINT, avg_player_value_eur BIGINT, max_player_value_eur BIGINT, days_since_last_valuation SMALLINT	Periodic snapshot, 1 row per team per month-end. squad_market_value_eur = sum of latest player valuations (semi-additive); days_since_last_valuation = freshness.
dw.dim_team	team_key INTEGER, tm_club_id INTEGER, team_name VARCHAR(80), fd_team_name VARCHAR(50), country VARCHAR(30), current_competition_key SMALLINT, coach_name VARCHAR(80), stadium_name VARCHAR(80), stadium_seats INTEGER, valid_from DATE, valid_to DATE, is_current BOOLEAN	SCD2. team_key surrogate (–1 unknown); tm_club_id natural key; fd_team_name = spelling in match files; coach_name, current_competition_key tracked Type 2; stadium_* Type 1; valid_to 9999-12-31 for the current row.
dw.dim_competition	competition_key SMALLINT, fd_division_code VARCHAR(4), tm_competition_id VARCHAR(4), competition_name VARCHAR(50), country VARCHAR(30), tier SMALLINT, teams_per_season SMALLINT	SCD1. Codes of both sources (E0↔GB1, SP1↔ES1, I1↔IT1, D1↔L1, F1↔FR1); teams_per_season drives completeness check.
dw.dim_season	season_key SMALLINT, season_code CHAR(4), season_label VARCHAR(7), start_date DATE, end_date DATE, is_complete BOOLEAN	Static. season_code as in source path ('2627'), season_label '2026/27', is_complete = all matches played.
dw.dim_date	date_key INTEGER, full_date DATE, year SMALLINT, quarter SMALLINT, month SMALLINT, month_name VARCHAR(9), iso_week SMALLINT, day_of_week SMALLINT, day_name VARCHAR(9), is_weekend BOOLEAN	Static calendar, date_key yyyyymmdd (–1 unknown), day_of_week ISO (1 = Monday).
dw.dim_referee	referee_key INTEGER, referee_name VARCHAR(60)	SCD1. Referee name as published (English leagues only).
staging.team_name_map	fd_team_name TEXT, tm_club_id INTEGER, match_method TEXT, match_score NUMERIC(4,3), reviewed BOOLEAN	Bridge football-data team name → Transfermarkt club_id; match_method exact/fuzzy/manual, match_score 0-1 for fuzzy.

7. Demo queries

All seven questions are answered in *sql/03_demo_queries.sql* (tested on PostgreSQL 16 with the sample data in *sql/02_sample_dml.sql*). A helper view *dw.v_team_match* unpivots each match into a home and an away row; it groups by the durable key tm_club_id so SCD2 versions roll up to one team. Example (Q2):

```
WITH fav AS (SELECT f.*, CASE WHEN odds_home <= LEAST(odds_draw, odds_away) THEN 'H'
                        WHEN odds_away <= LEAST(odds_home, odds_draw) THEN 'A' ELSE 'D' END AS favourite
            FROM dw.fact_match f WHERE odds_home IS NOT NULL)
SELECT c.competition_name, s.season_label, count(*) AS matches,
       round(100.0*avg((favourite = full_time_result)::int),1) AS favourite_hit_rate_pct
FROM fav f JOIN dw.dim_competition c USING (competition_key) JOIN dw.dim_season s USING (season_key)
GROUP BY 1, 2 ORDER BY favourite_hit_rate_pct;
```

Other queries: Q1 points per €100 m squad value per team/season; Q3 home/draw/away % and goal difference by league/season; Q4 goals per shot on target vs. points; Q5 cards per match by referee; Q6 points per match per coach version (SCD2); Q7 top-10 upsets by winner's odds.

Contribution and LLM disclosure

Member	Role	Contribution
Ron-Aran Paju	All roles (individual project): business analysis, data sourcing, architecture, modelling, SQL, report	100%

LLM use. Claude (Anthropic) was used to brainstorm the topic, check data source availability, draft the schema, SQL, diagrams and this report; all output was reviewed and edited by the author. Chat: [link to the shared Claude chat]